Your Project Title: A Descriptive Subtitle If You Need One*

Student One
sone@gatech.edu
Georgia Institute of Technology
Atlanta, Georgia, USA

Student Two
stwo@gatech.edu
Georgia Institute of Technology
Atlanta, Georgia, USA

Student Three
sthree@gatech.edu
Georgia Institute of Technology
Atlanta, Georgia, USA

Student Four
sfour@gatech.edu
Georgia Institute of Technology
Atlanta, Georgia, USA

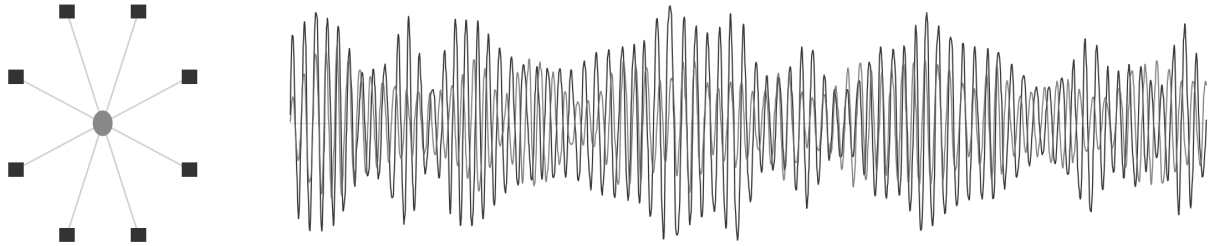


Figure 1: Optional teaser image spanning both columns — here, a spatial loudspeaker layout and a waveform. Replace it with a performance photo or interface shot, or delete the block entirely.

Abstract

Summarize your project in a short paragraph. State what you made — an instrument, a system, a piece, a performance — and the idea behind it. Briefly name the tools or techniques you used, describe how it was performed or tested, and end with what you learned or what the result demonstrates.

Keywords

live electronics, spatial audio, mapping, real-time synthesis

1 Introduction

Open with the idea. What is the project, and what motivated it? What question, sound, gesture, or performance situation were you exploring? Close this section with a one-sentence statement of what the paper contributes or demonstrates, so the reader knows where things are headed.

2 Background and Related Work

Situate your work. Mention the instruments, systems, pieces, composers, or research that your project builds on or reacts to. This is where you cite sources — for example, the laptop-orchestra tradition [3], foundational computer-music technique [1], or a tool you relied on [2]. Citations are numbered automatically; see the `L42i_paper_template.bib` file for how each entry is written.

*Prepared for *Technology Ensemble*, Fall 2026, Georgia Institute of Technology, School of Music.

3 System Design

Describe the architecture of what you built at a conceptual level, before the implementation details. What are the parts, and how does signal or information flow between them? A block diagram is usually the clearest way to show this (Figure 2).

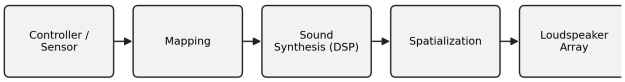


Figure 2: A signal-flow diagram of the system: input is captured, mapped to synthesis parameters, rendered as sound, spatialized, and sent to the loudspeakers. Replace this with a diagram of your own system.

4 Implementation

Now get concrete. What software and hardware did you use (for example Pure Data, SuperCollider, Max, sensors, microphones, a loudspeaker array)? How did you map input to sound? If the mapping or a piece of DSP is central to your project, show it. For instance, a simple amplitude control that scales a gesture value $g \in [0, 1]$ into a gain a might be written as

$$a = g^2, \tag{1}$$

which gives finer control at low levels. Refer back to equations by number, as in Equation 1.

5 Performance and Evaluation

Describe how the work was actually used — in rehearsal, in performance, or in testing. What happened? What worked well, and what did not? Concrete observations are more useful than general claims. A small table can help summarize results or configurations (Table 1).

Table 1: Example table: summarize a setup, a mapping, or results. Replace the rows with your own.

Element	Choice
Controller	Wireless motion sensor
Synthesis	Granular, real-time
Spatializer	8-channel VBAP
Latency	≈ 12 ms

6 Discussion

Step back and reflect. What did the project teach you about the idea you started with? What were the limitations, and what would you change? This is also the place to connect your specific result back to the broader themes of the course.

7 Conclusion and Future Work

Briefly restate what you did and what it showed, then point to where it could go next — extensions, a new piece, a more robust version, a different performance context.

Acknowledgments

Thank collaborators, performers, and anyone who helped with the project here. Delete this section if you don't need it.

References

[1] Miller Puckette. 2007. *The Theory and Technique of Electronic Music.* World Scientific Publishing, Singapore.

[2] The SuperCollider Community. 2023. SuperCollider: A Platform for Audio Synthesis and Algorithmic Composition. https://supercollider.github.io. Software.

[3] Dan Trueman, Perry Cook, Scott Smallwood, and Ge Wang. 2006. PLOrk: The Princeton Laptop Orchestra, Year 1. In *Proceedings of the International Computer Music Conference (ICMC).* International Computer Music Association, New Orleans, USA.